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# predict new data using model_7 on the prediction data created
preds_farc <- predict(model_7, newdata = prediction_data_farc, type = "response",
                      interval = "confidence")

# unlist predicted column
prediction_data_farc$predicted <- preds_farc[, "fit"]

# confidence intervals around predicted probabilities
prediction_data_farc$lower_95 <- preds_farc[, "lwr"]
prediction_data_farc$upper_95 <- preds_farc[, "upr"]

# plot predicted % Yes votes in the referendum against farc attacks
pdf("pred_plot.pdf")
pred_plot <- ggplot(prediction_data_farc, aes(x = exp(ln_atfarc_cede), y = predicted)) +
  geom_ribbon(aes(ymin=lower_95, ymax=upper_95), alpha = 0.2) +
  geom_point() +
  geom_rug(data = df, aes(x=exp_atfarc_cede),
          inherit.aes = FALSE, alpha = 0.4, sides = "b") +
  labs(y = "Predicted % Yes Votes in Referendum",
       x = "FARC Attacks",
       title = "FARC Attacks vs Predicted Votes in Referendum") +
  theme_minimal()
pred_plot
dev.off()

```